

Connecting brain and mind through temporo-spatial dynamics: Towards a theory of common currency

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ABSTRACT

Despite major progress in our understanding of the brain, the connection of neural and mental features, that is, brain and mind, remains yet elusive. In our 2020 target paper (“Is temporospatial dynamics the ‘common currency’ of brain and mind? Spatiotemporal Neuroscience”) we proposed the “Common currency hypothesis”: temporo-spatial dynamics are shared by neural and mental features, providing their connection. The current paper aims to further support and extend the original description of such common currency into a first outline of a “Common currency theory” (CCT) of neuro-mental relationship. First, we extend the range of examples to thoughts, meditation, depression and attention all lending support that temporal characteristics, (i.e. dynamics) are shared by both neural and mental features. Second, we now also show empirical examples of how spatial characteristics, i.e., topography, are shared by neural and mental features; this is illustrated by topographic reorganization of both neural and mental states in depression and meditation. Third, considering the neuro-mental connection in theoretical terms, we specify their relationship by distinct forms of temporospatial correspondences, ranging on a continuum from simple to complex. In conclusion, we extend our initial hypothesis about the key role of temporo-spatial dynamics in neuro-mental relationship into a first outline of an integrated mind-brain theory, the “Common currency theory” (CCT).

1. Introduction

1.1. “Common currency” of neural and mental features I – why we need it and what it is

One of the major scientific mysteries of our time is the connection of brain and mind. While this question has been debated traditionally as mind-brain problem in philosophy and, more recently as world-brain problem in neurophilosophy [49,50], neuroscience takes major empirical steps in investigating the neural basis of mental features like consciousness [55,72,77], self [51,66,78], thoughts [7,22], and many others. However, despite all progress in its empirical investigation as well as in the development of theoretical models of the brain, such as, to name a few examples, predictive coding/free energy principle [12], thermodynamic accounts of neural activity [31], the inside-out model [3], the distributive adaptive control model [87], cognitive ontology [65], and the intrinsic model [67], the exact relationship of neural and mental states remains yet elusive.

One major problem in these brain models is the lack of a direct relationship between neural states and mental features. The absence of such direct neuro-mental relationship means that the neural activity in these brain models could, in principle, operate in the same way without the presence of any mental states – which leads to a lack of direct connection between neural and mental states. Most of the current theories presuppose that some special feature in the brain, distinct from all others, accounts for the special nature of mental features. Candidates for such special features in the case of consciousness are, for instance, information integration [80], the global

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neuronal workspace [42], and various others [55]. In contrast, our [61] target article “Is temporospatial dynamics the common currency of brain and mind? Spatiotemporal Neuroscience” suggested a novel methodological strategy. Rather than searching for some special feature within the brain to account for mental features, we proposed that a most basic and unspecific feature of the brain’s neural activity is shared with mental features as their “common currency” providing their connection.

The idea behind the assumption of such common currency is simple. If two people from two countries would not share a basic and most unspecific feature, that is, a common language like English, they could not communicate and thus connect with each other at all in more specific ways. The same applies for a shared currency like the Euro in Europe, which as basic and unspecific feature, makes possible direct more specific connection among the different European countries’ economies. Searching for the common currency of neural and mental features may thus provide a first insight into their most basic and fundamental connection. We postulate that such unspecific feature serving as common currency provides the basis for a better understanding of the mechanisms underlying neuro-mental relationship.

1.2. “Common currency” of neural and mental features II – temporo-spatial dynamics as candidate

What is a candidate feature that could plausibly take on the role as “common currency” of neural and mental features? The answer in our 2020 paper [61] was clear and straightforward. As illustrated by various examples on self, consciousness, time perception and bipolar disorder, temporospatial dynamics is assumed to be shared by both neural and mental features as their “common currency”: the brain’s neuronal dynamics and topography are manifest on the mental level in corresponding subjective experiences of time-space, i.e., mental dynamics and topography (see below for definition).

Given that mental features and their neural basis are here explained by spatiotemporal features, we spoke of “Spatiotemporal Neuroscience” [62]. Roughly, Spatiotemporal Neuroscience aims to account for neural and mental features in primarily spatial and temporal terms rather than by cognitive, affective, sensory or motor functions, information integration, global neuronal workspace, prediction and free energy, (or other features). Thus, Spatiotemporal Neuroscience primarily focuses on the spatial and temporal features of the brain’s neural activity, that is, its dynamics and topography as these allow to connect and relate it to mental features [47].

Spatiotemporal Neuroscience does not only provide a novel empirical view of mental features but also a different methodological (or epistemological) approach to the question of mind-brain relation compared to most current neuroscience (and also philosophy of mind): it neither reduces or eliminates mental features to neural features nor does it consider them unrelated to neural features. Instead, Spatiotemporal Neuroscience searches for and identifies their common basis, that is, what they share as their “common currency” – which can therefore explain their connection and interdependence without reducing mental to neural features (as implied by reductive/eliminativist materialism and physicalism) or, conversely, reducing neural to mental features (as implied by idealism) [102].

1.3. From hypothesis to theory – “Common currency theory” (CCT) as first step towards an integrated brain-mind model

Four years have passed since our original paper in 2020. The current paper aims to develop the original paper in a threefold way by providing an empirical update, an empirical extension, and a first outline of a theoretical elaboration. First, we update our 2020 paper by extending the range of mental features lending support to temporo-spatial dynamics as common currency; we now consider other

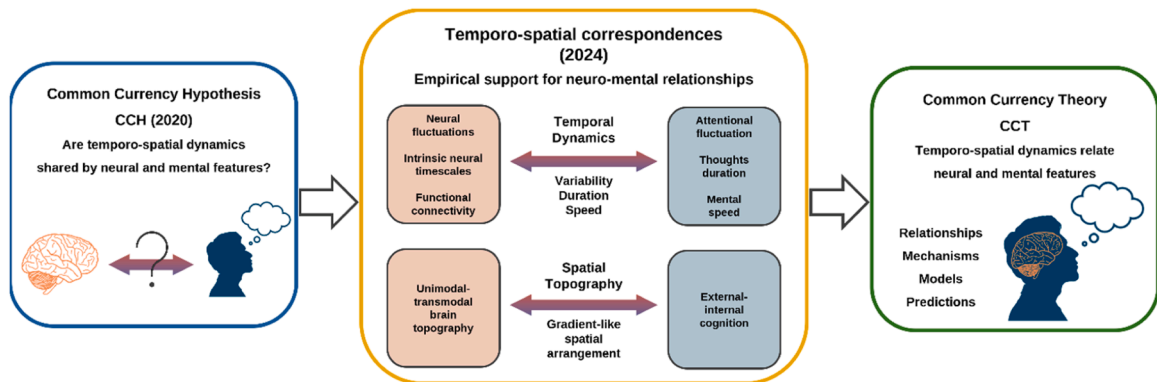


Fig. 1. Evolution from the Common Currency Hypothesis to the Common Currency Theory. The left box represents the initial 2020 Common Currency Hypothesis (CCH). The central panel depicts the proposed temporo-spatial correspondences between neural characteristics (orange boxes) such as neural fluctuations, timescales, functional connectivity, and brain topography, and, respectively, corresponding mental characteristics (blue boxes) such as attentional fluctuations, thought duration, mental speed, and external-internal cognition. The bidirectional correspondences form the “common currency” bridging brain and mind. The right box illustrates the expanded Common Currency Theory (CCT) with key features: systematic relationships, mechanisms, models, predictions. The arrows represent the theoretical progression from the initial CCH, through empirical support, to the comprehensive CCT framework.

examples like meditation, thoughts, attention, and depression. Secondly, unlike the original paper that focused primarily on the temporal domain, we now extend our common currency approach also to the spatial domain: we show how the brain's spatial features, that is, its topographic organization, are shared by neural and mental states. This will be illustrated by the mental effects of the brain's topographic reorganization in meditation and depression.

Third, we propose a first outline of a theoretical elaboration of the common currency. Specifically, we suggest a theoretical elaboration of the neuro-mental relationship by different forms of temporo-spatial correspondences, ranging on a continuum from simple to complex. We thus extend our initially proposed "Common currency hypothesis" [61] to a more comprehensive framework, leading to a "Common currency theory" (CCT). We are aware that this is only a first and incomplete outline of the CCT as an integrated brain-mind model; this will need to be elaborated in more formal/mathematical terms in the future (see for instance [58] for details). To clarify what we mean by temporo-spatial correspondence, we start the paper with its definition and the determination of its different forms – this will be illustrated by several empirical examples (Part I). That provides the theoretical background within which the empirical data on shared dynamics (Part II) and topography (Part III) as the common currency of neural and mental features can be viewed and understood (Fig. 1).

2. Part I: how are neural and mental features related? simple vs complex correspondence

2.1. Continuum of different neuro-mental correspondences – key role for time-space scales

Neural and mental features may be related in different ways, which may fall on a continuum of different relationships, ranging from weak to strong. At one extreme of the continuum we can find very strong relationships like numerical identity (A and B are the same and correspond in all aspects, being indeed the same) while, on the other hand, there are very weak relationships, such as correlation (when A changes in some aspects, B also changes in some respects). There is also a major middle ground from medium to strong relationships, such as isomorphism, resemblance, similarity, etc.... The key concept here is 'correspondence'. The concept of correspondence is an umbrella term which refers to a wide variety of relationships among two features, variables, or levels of description.

In the neuroscientific context, this amounts to a continuum of very weak neuro-mental correspondence at one end of the continuum to very strong ones at its other end. This continuum from weak to strong neuro-mental correspondences is understood in a purely empirical way, that is, it concerns the description of their relationship. In contrast, it does not refer to the nature of the relationship of neural and mental features, neither to its existence which is an ontological and thus more philosophical question [50]. We will therefore leave aside the theoretical issues concerning the nature (and existence) of this relationship (for instance: causality, emergence, identity, dualism, etc...). Instead, we are focusing solely on the description of the neuro-mental relationship: we search for the empirical identification of temporal and spatial dynamics shared by the neural and the mental, in terms of different degrees of correspondence. Such empirical (rather than ontological) focus implies that the concept of correspondence is applied here only to the temporo-spatial characteristics of neural and mental features, not to either brain or mind as such.

Presupposing such purely empirical context, we suppose that the degree of neuro-mental correspondence is related to the timescales on which neural and mental features operate. Roughly, timescales refer to the duration of a process, like the one second duration of a heartbeat or the 24 h duration of the day-night cycle. We now, analogously, assume that neural and mental features are related with each other through their corresponding timescales – such correspondence can take on different ways and degrees. If there is one-to-one correspondence in the temporal characteristics of neural and mental features, that is, in their timescales, one may speak of 'simple correspondence': the duration of the mental state is the same as the duration of the neural state thus involving only one timescale. If, in contrast, multiple temporal variables with different timescales are involved across neural and mental features, one may want to speak of 'complex correspondence': the durations of neural and mental states are different, which nevertheless does not prevent them from being related with each other in a more complex temporal way across their different timescales (as, for instance, in a temporally lagged way).

Taken in this sense, simple and complex correspondence of neural and mental features are extremes on a continuum of different temporal relationships with the involvement of single to multiple timescales. Since the same applies to the spatial features with their distinct space scales (different degrees of spatial extension), we speak of "temporo-spatial correspondence" which, as we assume, range on a continuum from simple to complex.

Note that the different forms of temporo-spatial correspondence along their simple to complex continuum neither entails (i) direct vs indirect neuro-mental relation, nor (ii) weak vs strong neuro-mental relation. (i) In both instances, simple and complex correspondence, there is a direct temporal relationship of neural and mental features, that is, through timescales including either single or multiple timescales – there is no intermediate or third (non-temporal) feature between the temporal features of neural states and those of mental features. The same kind of direct relationship of neural and mental features also applies to spatial features with their different space scales, that is, topography (see below). (ii) Nor does simple vs complex correspondence denote weak or strong neuro-mental relationship: simple neuro-mental correspondence through one single timescale can be both weak and strong in the same way their complex correspondence through multiple timescales can also be either weak or strong. Accordingly, the continuum from weak to strong temporo-spatial correspondences of neuro- mental relationships operates on a more fundamental and basic level than the higher-level distinctions of direct vs indirect and weak vs strong.

2.2. Simple correspondence: neural and mental features operate within a single (or limited number of) time-space scale(s)

We propose that the common currency can be characterized by such a continuum of different forms of correspondence of neural and

mental features, from simple to complex. There is for instance rather complex temporal correspondence with the involvement of multiple temporal features like different timescales between the temporal imprecision of the brain's phase cycles in schizophrenia subjects and their experience of temporal fragmentation [2,36,94]: the temporal imprecision of the phase cycles occurs on a microlevel in the lower millisecond range of 8 to 10 ms, while the experience of temporal fragmentation is manifest on a macrolevel, with a longer timescale of seconds or even minutes [29,88].

Instead, there seems to be a more simple form of temporal correspondence of both brain and experience when it comes to reduced speed/slowness in depression [48,53,56,74,75]: the decreased speed of the brain's neural activity (measured by neural variability) in regions which mediate inner time perception - such as the sensorimotor cortex - around 10 s are manifest in a corresponding abnormal slowness of the subjects' inner time perception in a similar time-scale around 10 s [56]. While the converse, high speed in the visual cortex leads to an experience of abnormally fast outer time speed in perception [56]. There is thus one-to-one correspondence of neural and mental features in their temporal (or spatial) structure which may be traced to the similarity in their respective timescales with the involvement of only one timescale.

How can we better account for the difference between the two examples of schizophrenia and depression in their cross-temporal neuro-mental relationships? We propose distinguishing the concepts of simple and complex correspondence. Let us first focus on simple correspondence. Simple correspondence refers to a one-to-one temporal relationship among neural and mental features with both operating in the same timescale or level. In one example, authors showed that both neural and sensory-cognitive attentional fluctuations hold in the same frequency range (0.01 to 0.1 Hz [99]). There is thus one-to-one temporal correspondence in the timescales of both neural and mental features. Analogous observations were obtained in the work on the 'Temporal Receptive Windows' by the group around Hasson [18]: they suggest one-to-one temporal correspondence between durations at the neural level and durations supporting cognition of words, sentences, paragraphs, etc.

Together, we assume similar timescales in both neural and mental features whilst characterizing their relationship by simple correspondence. In other words, a simple temporo-spatial correspondence can be understood in the sense that the neuro-mental relationship is characterized by the involvement of a single (or very low, and thus limited) number of time-space scale(s). Note that such simple temporo-spatial correspondence needs to be distinguished from correlation. Whereas classical correlation is based on the mapping of co-occurrences and localization of mental and neural events in the brain areas, correspondence is a stronger concept: it identifies similarity relations regarding the way specific neural and mental features are related to each other with respect to their temporal and spatial characteristics like similarity in their duration (or extension) within larger sets of neural and mental processes (e. g., perception and sensorimotor processing).

2.3. Complex correspondence I: neural and mental features operate across multiple time-space scales

This was the easy case. Not all neuro-mental relationships conform to such simple correspondence within a single (or limited number of) timescale(s), though. The relationships are often more complex entailing that temporo-spatial correspondence does not always mean proportional correlation between two variables in the sense of an isomorphism. Put in a more conceptual way, a good explanation does not necessarily require an isomorphic correspondence as the following: "A looks like B, so A explains B". For example, the dynamics in the horizontal motion of the car does not show an isomorphic correspondence with the dynamics in the process of combustion and vertical and circular motion of the engine; both are nevertheless related with each other as they exhibit some form of correspondence in their dynamics (see also [83]).

These more conceptual issues carry major implications for our characterization of the concept of correspondence. We need to account for those neuro-mental cases which exhibit some temporal (and/or spatial) connection across multiple time-space scales but do not follow the simple script of their correspondence within a single (or limited number of) time-space scale(s), though. We here speak of complex correspondence which shall be illustrated by some examples in the following.

Consider, for instance, subjects in sleep and anesthesia whose brain exhibits higher temporal variability in their Lempel-Ziv complexity (LZC) [103], that measures the degree to which the neural signal is compressed. Following the idea of simple correspondence, higher variability in the LZC means that multiple timescales are involved as the signal is more variable and therefore can take on multiple durations at single timepoints. Presupposing simple correspondence, one would then assume that the perception also shows multiple timescales with high variability: the degree of signal compression highly varies over time across different timescales (shorter and longer) in the brain's neural activity which should be manifest in an analogous variability of the perceived contents with distinct timescales, that is, durations on the mental level. That is not the case, though. Instead, we can safely assume that especially the subjects in general anesthesia do not perceive anything at all, there are no contents any more let alone changing durations in their perceived contents [103]. One can thus no longer speak of simple correspondence with a one-to-one neuro-mental relationship but rather of a more complex temporal relationship with different numbers of timescales, that is, multiple on the neural, and single or better none at all on the mental level.

But how is such an opposite dynamic pattern in the timescales of brain and experience generated? We assume that the higher variability of the LZC in the brain's neural activity relates to its higher degree of randomness during anesthesia, with a very high number of different timescales: spontaneous neural activity is here no longer constrained by the scale-free relationships of the awake brain in both its topography [79] and dynamics – this makes possible the dynamic exploration of a very high number of extreme timescales that are usually constrained and suppressed by the brain's scale-free dynamical organization [23,30,59,101]. Such unconstrained neural variability with the involvement of an extremely high range of different timescales, in turn, render impossible the structured perception and experience of any content – the extremely high neural variability with the various timescales may then be manifest in an opposite dynamic on the mental level, low or absent variability with only a single or even no timescale at all anymore.

Together, we consider the concept of correspondence to serve as umbrella term for different temporo-spatial relationships of neural and mental features, along a continuum of single to multiple time-space scales. We therefore speak of temporo-spatial correspondence: this can be characterized by a continuum from single to multiple time-space scales across neural and mental features through which their relationship is constituted in primarily temporal and spatial terms. To understand this better, let us describe some other empirical examples.

2.4. Complex correspondence II: involvement of multiple time-space scales across neural and mental features – empirical examples

Complex correspondence means that neuro-mental relationships may operate across different time-space scales, while nevertheless sharing a certain more basic underlying dynamic feature like speed, fluctuation or duration. For instance, [1] showed that neural variability in the infraslow frequency range relates directly to behavioral variability in a faster frequency range (0.8 to 1 Hz), during a sustained attention task. Given that the longer timescale of neural activity relates here to a shorter timescale on the cognitive-behavioral level of attention, there is no longer simple correspondence with the involvement of only one timescale. While this example does not directly refer to experience itself, it nevertheless illustrates one example of a complex temporal relationship of neural and cognitive levels across different timescales.

This is further supported by other examples showing the reverse direction in the mismatch of neural and mental features in their timescales. For instance, internal and external thoughts last about 10 to 30 s [69,84] which, on the neural level, can be measured and distinguished from each other in the much shorter timescales of theta (5 to 8 Hz), and alpha frequencies (8 to 13 Hz) [22]. Analogously, the timescale of deidentification with the own thoughts in meditation, that is, the subject's non-attachment and non-self-specific relation to the contents of its own thoughts, is a process that lasts at least several seconds on the mental level [68], while on the neural level, thought deidentification can be measured on the much shorter timescales of the EEG (1s to 100 ms) in their respective durations, indexed by its autocorrelation window length [68].

Yet another example of the involvement of multiple timescales can be found in schizophrenia. Schizophrenia can be characterized by increased and highly irregular phase shifts in its neural activity in the lower millisecond range around 10 ms [36,93,94]. This very short timescale cannot be accessed in our subjective experience or consciousness which lasts over several seconds [63]. Does this mean that these irregular phase shifts are not experienced at all by the schizophrenic patients? There is initial evidence that these phase shifts around 8 to 10 ms are experienced on the longer timescales of consciousness, that is, in 30 ms to 50 ms or longer timescales of one or several seconds where they are manifest in the experience of 'temporal fragmentation' [2,35,76].

Together, these examples show the involvement of multiple timescales in the relationship of neural and mental features which, though not sharing one and the same timescale, may nevertheless be related with each other across their multiple timescales. This no longer conforms to 'simple correspondence' as outlined above but is well in line with what we describe as 'complex correspondence'. As illustrated by our different examples, complex correspondence may include a variety of distinct neuro-mental relationships with all operating across different timescales (and space scales).

Accordingly, complex correspondence is distinguished from simple correspondence by the involvement of multiple (rather than single) time-space scales which, mutually influencing each other, characterize neuro-mental relationships in a complex (rather than simple) temporo-spatial way. This entails that the concept of the common currency may be sketched in slightly distinct ways in simple and complex correspondence: the common currency consists in a single and specific timescale shared by both neural and mental features in the case of simple correspondence. While, in the case of complex correspondence, neural and mental features no longer

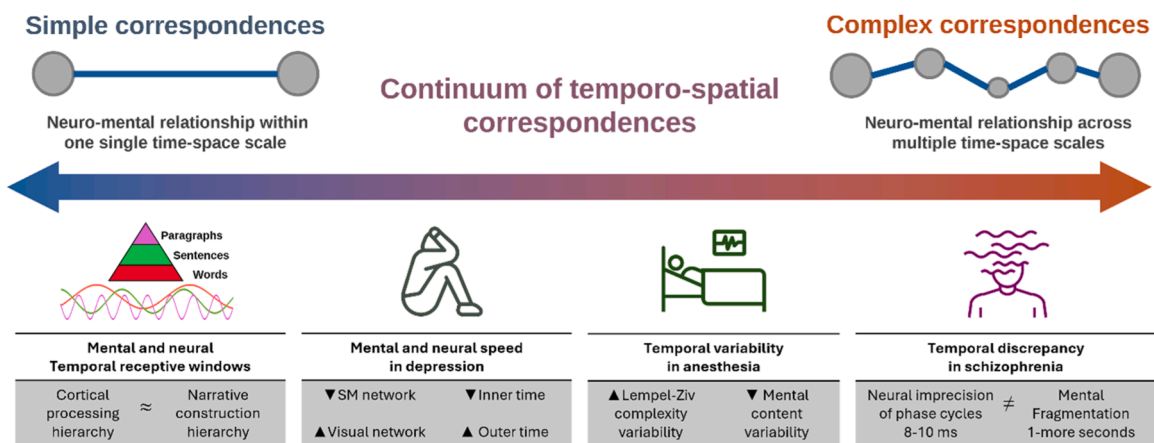


Fig. 2. Continuum of temporo-spatial correspondences between neural and mental features. Simple correspondences (left) involve neuro-mental relationships within a single time-space scale. Complex correspondences (right) represent neuro-mental relationships across multiple time-space scales. Examples along the continuum illustrate cases ranging from simple (e.g., cortical and narrative temporal receptive windows) to complex correspondences (e.g., temporal discrepancy in schizophrenia), with some middle cases (e.g., neuro-mental speed in depression and temporal variability in anesthesia).

share one single specific timescale, that is, the same, but are related with each other across multiple time-space scales across the continuum of shorter and longer timescales (and less and more extended space scales). Accordingly, the common currency may then be determined on a continuum from single to multiple scales of time and space characterizing the neuro-mental relationship (Fig. 2). That shall be illustrated in more detail in the following empirical parts focusing on temporal and spatial correspondence.

3. Part II: temporal correspondence - Dynamics is shared by neural and mental features as their “Common currency”

The assumption of neuro-mental correspondence provides the theoretical background within which the common currency can be viewed and understood. This shall now be substantiated by various empirical examples of dynamics being shared as common currency by both neural and mental features.

Dynamics refers to the temporal dimension describing the pattern of changes over time. Taken in this sense, dynamics can include different dimensions like variability, timescales, scale-free dynamics and synchrony [47]. Specifically, there is the dimension of duration, that is, how long a process lasts: its dominant timescale. For instance, the timescale or duration of a heartbeat is one second while the duration of a day is 24 h. Analogously, the brain’s neural activity includes different durations, which are described as intrinsic neural timescales [16,92]. Yet another feature of the brain’s dynamic are its fluctuations: there is constant change in the amplitude of the neural activity, as it can be measured by neural variability and entropy [90]. Finally, there is the speed of neural activity which may operate in a slower or faster mode. We now show that these three dynamic dimensions are manifest in the connection of both neural dynamics and mental dynamics as their “common currency” – we will specify that connection by what we introduced as “correspondence” in the first part.

3.1. *Fluctuations – variability of neural activity shapes variability of attention and thoughts*

The brain’s neural activity is never the same as it fluctuates over time. Its amplitude changes from time point to time point as it can be measured by its neural variability, that is, the root means square of its amplitude [10,90]. Various studies observed that the brain’s neural variability is manifest in the variability of attention and thoughts – that suggests some degree of their temporo-spatial correspondence.

A recent fMRI study [99] demonstrated that the neural variability of specifically the default mode network (but not the one of the dorsal attention network) is related to the variability of the reaction time during a sustained attention task in both auditory and visual modalities (see also [96]). Moreover, even more intriguing, variability in both neural and behavioral activity operated in exactly the same frequency range, (0.01 and 0.1 Hz), as both neural and behavioral data were band-passed in similar ways [99]. This suggests direct temporal relationship of brain and attention-related behavior (reaction times, RT) through their variability in shared frequencies (see also [1] for similar findings of direct connection of neural and behavioral variability).

This is further supported by other studies. [34] conducted a unique experimental approach in that they recorded the variability or fluctuations in all three measures including behavior (reaction time in response during a continuous performance task (CPT)), mind-wandering (off-thoughts during the CPT task), and DMN activity (change in amplitude during performing the same trials and cognitive demands in the CPT task) (see also [73] for an overview of such multi-level studies). They show significant relationship of the percent signal change in the task-related amplitude of various DMN regions with both off-task mind wandering rating and RT variance: the greater the DMN amplitude variability, the larger the degree of off-thoughts (as in the visual analog ratings), and the lower the RT variance during the attention task (see also [99] for a similar relationship).

Similarly, fluctuations or variability in dynamic functional connectivity inside and outside the DMN have been related to spontaneous thought like off-thought [6,7,32,33,43,73,91,98]. Together, these findings strongly suggest that variability of neural activity in the default mode network corresponds to the variability of attention and mind wandering. Hence, fluctuations and, more specifically, variability may be shared by neural and cognitive activity as their “common currency”. The empirical data suggest, in part, simple correspondence, that is, neural and cognitive features fluctuate within one and the same time scale as it seems to be the case in attention. While other data like the ones on thoughts hint towards more complex correspondence with fluctuations across different timescales.

3.2. *Duration – timescales of the brain relate to the timescales of stimuli and thoughts*

Duration is yet another feature of dynamics. Duration refers to the time for which some event takes time to be processed and completed like the 1 s of a heartbeat and the 24 h of day-night; as such, duration is closely linked with the notion of timescales. Timescales refer to the length of a particular process. Various data show that the brain’s Intrinsic Neural timescales (INTs) relate to corresponding timescales in both external stimuli and internal thoughts. Let us start with the external stimuli.

The group around Hasson demonstrate that the brain’s neural activity exhibits different durations in its temporal windows through which it receives and processes external stimuli, i.e., temporal receptive windows (TRW) [17,97]. For instance, single words are mediated by shorter TRWs in the sensory regions while sentences or even whole paragraphs are related to longer TRWs in higher order regions [5,6,37]. This suggests direct one-to-one temporal correspondence with simple correspondence between the durations of the external stimuli and the durations of the neural activity - duration thus serves as their common currency [16,92].

What holds for external stimuli also holds for internal thoughts. Different kinds of thoughts, i.e. thought dimensions, show different durations and thus distinct timescales. For instance, on-task thoughts, that are related to task one performs, are shorter and are mediated by the alpha frequency (defined as the frequency range between 8 and 13 Hz), while off-task thoughts, that remain unrelated

to the task at hand, are longer and are related to the slower theta frequency (5 to 8 Hz) [22]. Similarly, internal thoughts, as related to the own self, show longer duration than external thoughts that are about the outer environment: internal thoughts are mediated by longer timescales in both behavior and neural activity while external thoughts relate rather to shorter timescales like faster behavioral and neural activities [104]. Hence, there seems to be a connection between the timescales of the brain's neural activity and those of thoughts – the timescales may thus provide their common currency. We need to be careful, though as we have only indirect access to the thoughts through the subjects' reporting – we can therefore not properly distinguish between the effects of the thought themselves and those of our reporting [82].

The connection of the timescales of both neural activity and thought is further substantiated by directly measuring the duration of neural activity through its similarity with itself over time, i.e. the autocorrelation window (ACW) [15,20,21]. How are the brain's timescales and its ACW related to mental features? This was probed in a recent EEG study that compared different forms of meditation that can be distinguished by the width of attentional focus required [86]. Specifically, a narrow focus of attention on a temporally segregated sequence of different body parts as required during body scan meditation exhibits short ACW. While the ACW was rather long in the case of Shoonya meditation, where subjects are supposed to widen their attentional focus without paying attention to any specific content. Accordingly, the width of the attentional focus during the different forms of meditation relates to the width of the processing windows in the brain's neural activity, measured as the duration of its ACW [86]. Together, these findings suggest that duration and its timescales are shared by both neural activity and the width of attention as their "common currency" along a continuum ranging from simple to complex forms of temporo-spatial correspondence.

3.3. Speed - Abnormal slowness and fastness in the neural and mental activity of depression

Speed is a basic dimension of time; it describes how much distance is covered by something over a certain unit of time. This also applies to the brain as its neural activity may advance over time with a certain speed. Speed can, for instance, be measured by the degree of change of neural activity: if there a lot of change in the amplitude or the phase of the signal over a short period of time, there must be high speed in the neural activity as it can be measured by the variability of its amplitude [56] and phase (through frequency sliding) [1,22]. One can thus speak of 'neural speed' (of amplitude or phase). Importantly, the brain's neural speed seems to have a counterpart on the mental level: mental speed. This can, for instance, be observed in major depressive disorder (MDD), where the variability and thus the speed of the subjects' mental activity is abnormally slow like when they think always about one and the same thing, issue or content, e.g., rumination [47,53]. Such abnormally slow speed is not only manifest in the thoughts of depressive subjects but also in their other psychological functions like time perception, emotion, cognition, and even movements and visual perception; that shall be shown in the following.

Subjects suffering from depression often show abnormally slow movements which is referred to as psychomotor retardation [75]. Recent fMRI studies showed that neural activity from the brain's global cortical activity, including the visual cortex as input region to the motor cortex, is indeed too slow: their functional connectivity is delayed in MDD subjects with psychomotor retardation while it is too early in agitated anxious MDD subjects, resulting in psychomotor agitation [38,74,75]. Accordingly, the speed of the brain's neural activity in its global to motor cortex functional connectivity relates to the speed in the subjects' behavior, that is, their psychomotor retardation or agitation.

Depressed patients also experience abnormal slowness in inner time perception while their outer time perception is too fast. We therefore investigated speed on the neural level by measuring the degree of change, i.e. neural variability, depressed and manic bipolar patients, that is, we search for the neural variability changes related to inner time perception and those associated with outer time perception [56]. We observed decreased neural variability in the sensorimotor cortex, which is thought to mediate inner time perception, while neural variability was increased in visual cortex that mediates outer time perception in depressed bipolar patients – they experience their inner time speed as too slow while they perceive the outer time as too fast [56]. The converse neural pattern holds in manic patients who experience their inner time as too fast while outer time is too slow for them.

Together, these results suggest close relationships ranging from simple to complex temporal correspondence of neural and mental speed in both inner and outer time respectively. Speed also features in our thoughts. William James already pointed out that our thoughts come and go like the perchings of birds ([26], p. 18) which can occur in different speeds. Subjects suffering from depression often experience and report abnormally slow flow and speed in their thoughts [13,76]. Such abnormal slowness in the speed of thought can be investigated in more quantitative ways. For instance, [69] measured the degree of change and awareness of internally and externally oriented thoughts in MDD subjects. This yielded a combined time series of thought change and awareness which can be transformed into a power spectrum. The MDD subjects showed a lower power in their power spectrum and slower frequency of thought change than the healthy subjects which also correlated with the severity of rumination as measured by a rumination scale [69]. The thoughts in MDD can thus be characterized literally as slow and powerless.

One may now ask whether such abnormally slow speed of thought is related to an abnormally slow speed of neural activity. While such direct combined speed investigation of both thought and brain is still pending, there is indeed increasing evidence for an abnormally slow speed in the brain's spontaneous activity in depression: depression can thus be conceptually framed as a "speed disorder" ([47]; Scalabrini et al., 2024). In (Scalabrini et al., 2024), authors directly measured the speed in different frequency ranges of fMRI and found decreased power in mainly the infraslow frequencies in visual cortex, motor cortex, medial prefrontal cortex and subcortical regions such as hippocampus and amygdala. Most interestingly, such decreased speed, measured by shifts of median frequency (MF) towards the slower end of the power spectrum, correlated with the severity of both overall depression and psychomotor retardation with slower neuronal speed leading to higher symptom severity.

Moreover, MDD subjects show decreased fMRI neural activity in response to fast visual stimuli, further substantiating the speed

deficit hypothesis at the neural level [39]. Finally, [35] investigated the phase cycles in the theta and alpha range in their change over time - i.e. speed - during the resting state. MDD subjects exhibited lower variability in their alpha phase cycles than healthy subjects and, even more important, their phase cycles were always delayed, that is, they lagged their supposed ‘ideal’ timing. This suggests that the abnormal slowness in MDD operates across different timescales, that is, speed is affected in both shorter and longer timescales. This, as the example of depression shows, applies to both neural and mental activity who thus seem to share their abnormal slowness with decreased speed as “common currency” on a continuum from simple to complex temporal correspondence (Fig. 3).

4. Part III: spatial correspondence - Topography is shared by neural and mental features as their “Common currency”

4.1. Topography – spatial organization with shared gradients on neural and mental levels

The concept of *topography* comes originally from geography, where it refers to the organization and structure of natural features like mountains, lakes, forests, etc. including how they relate to each other. We here take into view the brain’s topography: the macroscopic spatial arrangement of various cortical features and networks across the brain surface, which allows to describe the cortical landscape with a parsimonious series of spatial trends of organization along hierarchies (see below for the concept of hierarchy) [19]. We will see, for instance, an example of global topographic organization in the spatial gradients displayed along the unimodal and transmodal regions, shaping what has also been described as a “core-periphery organisation” [30,40]. While examples of the brain’s more local topographies concern the organisation into lower- and higher-order sensory regions [4] as well as the three-layer topographic organisation of self with interoceptive, exteroceptive and mental layers as mediated by corresponding regions related to the aforementioned processes [66].

One may now raise the question of how the brain’s spatial organization, i.e. its topography, relates to the mental level. This leads us to the concept of ‘mental topography’ which, roughly, refers to the systematic organization and hierarchical relationships among the various mental contents within our experience. We propose that, just as the brain’s regions are organized in a systematic topographic way, the various contents in our experience are also organized in a spatial way along a systematic topography – a mental landscape with a structured organization of internal and external contents [60]. For instance, we will see that the gradient of the unimodal-transmodal topography of the brain relates to our experience of the relation between our internal self and the external

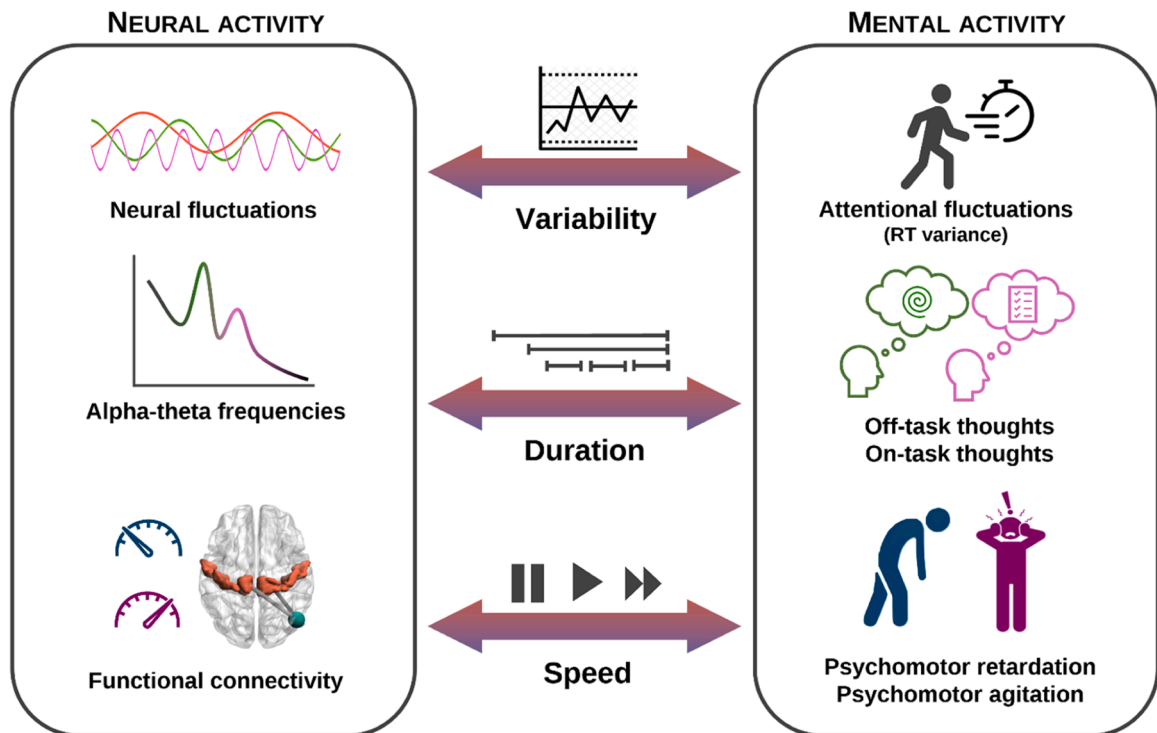


Fig. 3. Shared temporal dynamics as correspondences between neural and mental features. The bidirectional arrows depict the proposed spatio-temporal correspondences, where shared temporal dynamics – variability, duration, and speed – across neural and mental domains serve as the “common currency”. As examples, these dynamics include neural fluctuations, intrinsic neural timescales, functional connectivity, finding parallels in cognitive-behavioral variability, thought durations, and mental speed. Neural fluctuations exhibit a temporal correspondence with attentional fluctuations manifested as reaction time variance. The durations of alpha and theta bands also map onto the respective durations of on-task and off-task thought contents. Furthermore, patterns of functional connectivity, reflecting the speed of information flow, correspond to the speed of psychomotor behavior, with abnormalities linked to psychomotor retardation or agitation.

environment in a gradient-like spatial arrangement, a particular ‘mental topography’.

We shall remark that the concept of mental topography does not entail any ‘mapping’ of mental on neuronal features nor the localization of the latter in the regions of the former; this diverges, for instance, from the notion of isomorphism as described in (O’Regan, 1992), as we do not invoke representational correspondences (scale models) between mental states and brain activity. Instead, we assume correspondence as defined above: the systematic organization and hierarchical relationships among the different regions/networks in the brain share their topography with the organization and hierarchy among the elements of the mental features, that is, the contents we perceive, associated with our self, and dominate our consciousness and other cognitive processes. Accordingly, we replace the concepts of ‘mapping’ and ‘localization’ by the one of correspondence with its continuum from simple to complex. In this sense, as neuro-mental correspondences at the temporal level were presented above, correspondences at the spatial level can similarly be identified. Despite their different elements (cortical regions and contents), the brain’s neuronal topography can then be related to the mental features and their topography, both corresponding to each other in their systematic organization and hierarchical relationships.

4.2. Neuronal topography shapes cognition – current literature

How does the concept of mental topography as advanced in this article relate to the current body of literature? It is known that regions that support cognition in the absence of external input, particularly self-generated cognitive processes, are located at the extreme end of the macroscale gradient [14,85]. Notably, regions like the Default Mode Network (DMN), traditionally considered task-negative, are now recognized for their active role in facilitating complex cognitive functions even when external stimuli are absent [40]. For instance, it has been shown that when a task demands decision-making in the absence of support from the external environment, that is, an internally-guided decision making [45,46] can be observed, featured by a shift in neural activity towards transmodal areas [44]; this finding, Murphy and colleagues argue, suggest that the topographic distance from primary sensory regions, which indexes the progressive independence of neural activity from environmental input, supports the progressive generation of complex cognitive states independent of environmental input.

This “shift” that co-occurs with cognitive tasks of increasing richness, which we observe when moving away from unimodal to transmodal regions, can be arguably thought as a rebalancing of internally/externally oriented mental states, very similarly to what we proposed with the concept of mental topography. Therefore, we do not present the concept of mental topography as separate from current neuroscience, but rather as a complementary idea that connects and bridges the gap between two distinct fields of research.

Moreover, it has been suggested that internally directed cognition (IDC) and externally directed cognition (EDC) do not merely compete for limited resources but can also co-occur [11]. Building on this argument, mental topography offers a potential explanatory framework for the possibility, proposed by the same authors of the aforementioned theoretical framework, of cooperation between these two modes of cognition. Mental topography not only provides a parsimonious dimension of variation to describe the gradual co-occurrence of IDC and EDC, but also accounts for the potential for their cooperation. This cooperation may occur in mental states—such as those involved in our cognition in general and may be changed in depression and meditative practices, which we will explore in the following sections.

4.3. Dual topography in brain and cognition – from the unimodal-transmodal gradient to the external-internal gradient

The brain’s spontaneous activity shows a peculiar spatial topographic organization, with a gradient (or continuum) from unimodal sensory to transmodal cognitive regions [30,40]. At one end of this gradient, there are unimodal regions like visual and auditory cortex and motor cortex; at the other end of the gradient, there are transmodal regions, like the prefrontal cortex and the default mode network regions [24]. On one hand, unimodal regions are more related with externally-oriented cognition such as perception, on-task thoughts, and externally oriented thoughts about the external environment [22,64,69,73,84].

On the other hand, transmodal regions mediate internally-oriented cognitive processes such as self-referential processing [51,52,78], off-task thoughts [22], internally oriented thoughts [69], and episodic simulation of past and future [71]. Accordingly, there seems to be a direct relationship between the brain’s neuronal uni-transmodal topographic organization and the mental organization of the directedness of thoughts, (that is, internally- and externally-oriented thoughts).

Interestingly, this cortical gradient also relates to analogous gradients in structural features, like myelin density [25], and temporal features like INTs [5,15]. Transmodal regions show lower myelin density and longer INTs with respect to unimodal regions, whose myelin density is higher, and INTs are shorter. This amounts to an intrinsic hierarchical organization within the brain’s spontaneous activity, where the notion of hierarchy refers to “a gradient of structural or functional features, that is, an orderly spatial progression of parameters which are changing systematically across the cortical sheet” [19].

How is such hierarchical organization along a uni-transmodal gradient related to our cognition? We have previously suggested a ‘Baseline model of cognition’, where the regions devoted to either internal or external cognitive processes are distinguished by both specific locations in the hierarchy (unimodal vs transmodal) and different intrinsic timescales duration (short vs long) [60]. Importantly, there is a gradient between the internal oriented cognition of our self on the one hand and the external oriented cognition of the environment on the other. Hence, the brain’s dual organization of unimodal-transmodal regions shows some degree of correspondence with the dual “topographic” organization of cognition, that is, internal and external.

Taken together, this “dual” topography may be shared by both neural and mental levels as their “common currency”: the dual uni-transmodal topographic organization on the neural level of the brain is related to (and ultimately manifest) in a corresponding dual spatial-topographic organization of internal and external cognition on the mental level. This is further supported by empirical

observations that shifts in the brain's unimodal-transmodal gradient lead to corresponding topographic changes on the mental level, that is, in the balance or gradient of internal and external cognition of self and environment. That shall be the focus of the following section.

4.4. Topographic reorganization of neural and mental states i – depression and mania

MDD can not only be characterized by abnormal slowing down of the brain's activity, as discussed in the previous chapter, but also by changes in its unimodal-transmodal topography. There is strong evidence that the unimodal-transmodal gradient is shifted towards its transmodal end at the expense of the unimodal regions. In MDD, recent evidence suggests an increased influence of transmodal regions on lower-order unimodal regions, which become more synchronized and thus "subjugated" to activity in the higher-order areas of the brain [53,70]. This is also manifest in increased global activity - indexed by GSCORR (Global Signal Correlation) - in transmodal regions such as the DMN, while it is decreased in unimodal regions like the visual cortex [70]. Together, these findings demonstrate that the unimodal-transmodal gradient exhibits an extreme disbalance in MDD with an abnormally strong shift towards the transmodal pole in its topographic organization.

Such extreme shift towards the transmodal regions is also manifest on the cognitive or mental level. Depressed subjects show typically abnormal increase in internal cognition and attention with a strong focus on their own thoughts (rumination) [69], self (increased self-focus) [53], and on the past, at the expense of the future [53,89]. While, at the same time, external cognition and perception are decreased, which is manifest in decreased sensory unimodal activity the visual cortex [74,75].

Most interestingly, an analogous but opposite shift can be observed in mania as in bipolar disorder. Here, the brain's topography shifts away from the transmodal regions towards the unimodal regions, displaying increased activity in unimodal regions, like in motor and visual cortices, and concurrently decreased activity in transmodal regions, such as the DMN [8,41,56,100]. While an analogous shift can be observed on the cognitive level with abnormally strong external cognition, together with a decrease in internal cognition [8,56].

We see that, similarly to depression - although in a converse way - mania again illustrates how shifts in the brain's unimodal-transmodal topography are manifest in corresponding shifts of external and internal cognition. The brain's neural topography with its unimodal-transmodal re-organization towards its extreme ends (either unimodal or transmodal) relates to corresponding shifts towards either internal or external cognition, that is, mental topography. This illustrates that, as said above, mental topography refers to the organization of the relationships between the different elements of our cognition (such as internally and externally directed cognitive processes).

Together, the topographic reorganization towards its extreme ends in depression and mania reinforces the notion of a dual topographic organisation of mental contents into two forms of directedness of mental contents, that is, internal and external contents. Dual topography may thus be "common currency" of neural and mental levels: it surfaces in a similar dual way on neuronal and mental levels while being manifest in distinct gestalts, that is, in the unimodal-transmodal gradient and the internal-external cognition gradient. Given its extreme shifts in both depression and mania, we propose that such 'topographic reorganization' is shared by both neural and mental levels as their common currency [53].

4.5. Topographic reorganization of neural and mental states ii – meditation

Yet another example of a shared topographic reorganization on both neural and mental levels can be found in meditative practices, which are often featured by the experience of a dissolution of the self. One term describing this phenomenon is 'non-dual awareness', as often reported by highly proficient meditation practitioners [27]. Roughly, non-dual awareness is defined as the experience of 'pure consciousness' beyond the duality of self and other/environment; this pertains to a deeper layer of our consciousness, the "silent nature of pure consciousness" [81].

Such non-dual awareness is well reflected in the following description: "an open, awake cognizance that precedes conceptualization and intention, and contextualizes and unifies both extrinsic task-positive and intrinsic self-referential mental processes, without fragmenting the field of experience into opposing dualities" [28]. Non-dual awareness is thus characterized by a profound reorganization of the distinct elements within our experience itself, as in that state, one no longer experiences a separateness between their personal sense of self and the other/environment, rather a sense of boundless unity [9].

The direct experience of non-dual awareness has often led to the assumption that meditation leads to a loss of the self altogether [9]. However, in line with above all Eastern traditions, we suggest that what is lost is only the duality or, put more extreme, the opposition between self and other/world in our experience – this is described by the concept of non-dual awareness. Together, these observations suggest that non-dual awareness reflects a profound topographic reorganization of the relationship of self and world in our experience, which are reported to be no longer experienced as distinct or dual but instead as integrated and nondual.

What are the neural substrates of non-dual awareness? To investigate the brain's spatial topography, we reviewed brain imaging findings (fMRI) during different types and experience levels of meditation including studies probing the interaction of meditation and self [9]. The main results are: (i) reduction in default-mode network activity (DMN) in experienced meditators during both meditation alone and self x meditation interaction; (ii) the DMN shows an increasingly positive correlation with the central executive network (CEN) in advanced meditators. These findings suggest a profound organizational shift of the brain's spatial topography in advanced meditators – we therefore proposed a "Topographic-dynamic Re-organization model of Meditation" (TRoM).

One core component of the TRoM is that the topographic re-organization of DMN and CEN is related to a decrease in the 'mental self' (or narrative self) with a relative increase of a more basic sense of a non-dual self, the minimal or experienced self - this is

experienced as non-dual awareness, i.e., non-duality of self-other/world. Accordingly, the experience of non-duality is related to an analogous non-dual topographic organization of the brain's neural activity – non-dual topography is shared on neural and mental levels as their “common currency” [61]. These findings further support our assumption of (i) the concept of mental topography referring to the relationship among different elements in our cognition (like internal self and external environment); (ii) its close relationship to the brain's neural topography, i.e. unimodal-transmodal gradient and DMN—CEN, and (iii) corresponding pattern of topographic re-organization on both neural and mental levels in depression, mania and meditation (Fig. 4).

5. Conclusion – from hypothesis to theory

5.1. Substantiating and extending the hypothesis of common currency

The relationship between neural and mental states remains yet unclear. Current theories often assume a special feature in the brain's neural activity to mediate mental features like consciousness, thoughts, self, etc.. We departed from that in our 2020 paper [61] by assuming a key role for a non-special feature in the brain's neural activity, that is, its temporo-spatial dynamics in providing direct connection of neural and mental features. Temporo-spatial dynamics provides such direct connection as it is supposed to be shared by both neural and mental features; this argument resulted in the “Common currency hypothesis (CCH)” [47,61], which was supported by various empirical examples available at that time.

We here further substantiate and extend the CCH in three respects. First, we extend the range of mental phenomena to which the CCH applies by providing novel examples like thoughts, attention, depression and meditation. Secondly, even more important, we here show for the first time how the brain's spatial features, that is, its topography, can provide a shared feature on both neural and mental levels, that is, how neural topography is manifest in what can be described as ‘mental topography’. Third, we characterize the neuro-mental relationship by temporo-spatial correspondence with a continuum from weak to strong similarity within or across different time (and space) scales, that is, simple to complex correspondence.

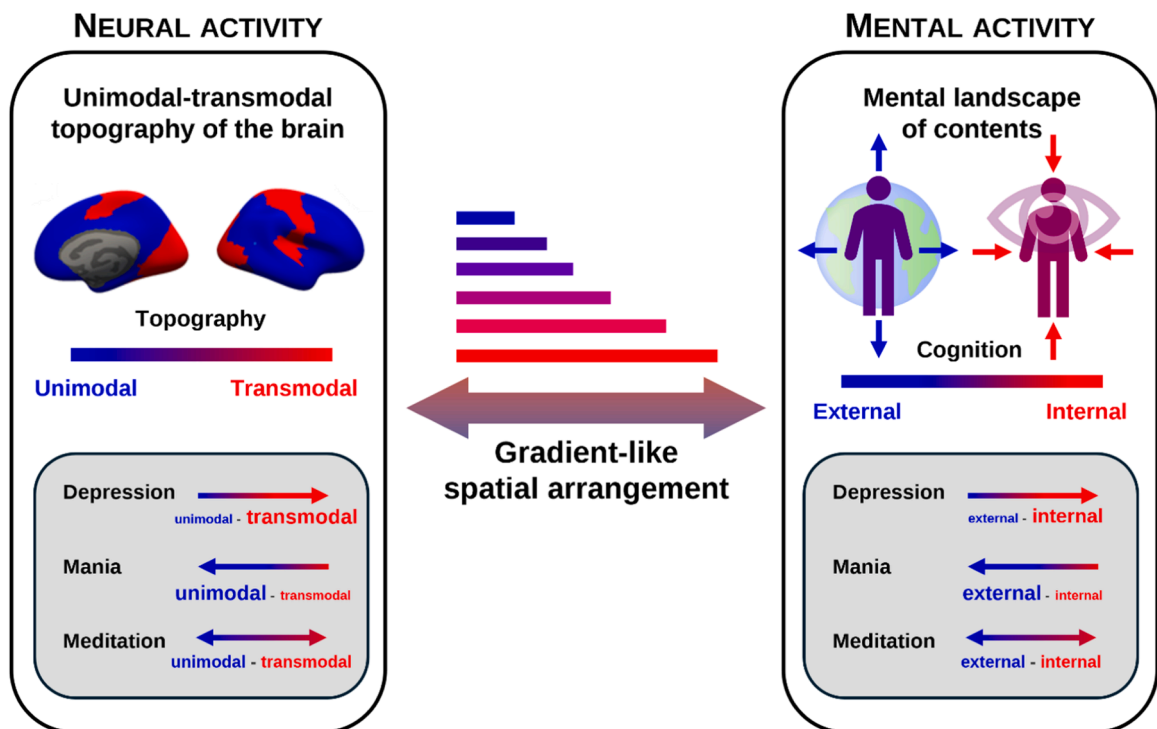


Fig. 4. Shared Topographic organization as correspondence between neural and mental features. The figure illustrates the correspondence between the organization of neural topography and mental topography and its reorganization in various illustrative conditions. On the left, the unimodal-transmodal gradient represents the spatial organization of brain regions, from unimodal (e.g., sensory) to transmodal (e.g., default mode network) areas. This neural topographic gradient corresponds to the mental topography of external-internal cognition, depicted by the icons of an externally-oriented and internally-oriented figure. The arrows indicate that conditions like depression, mania, and meditation involve reorganization along both neural and mental topographic gradients. In depression, there is a shift towards the transmodal end of the neural gradient and the internally-oriented mental gradient. Mania exhibits the opposite pattern, with a shift towards unimodal neural regions and externally-oriented cognition. Meditation is associated with a dynamic and balanced reorganization across the unimodal-transmodal and external-internal gradients.

5.2. Shifting from “Common currency hypothesis” (CCH) to “Common currency theory” (CCT) i – temporo-spatial relationships and mechanisms

Together, we here extend the CCH into the level of theory, a first outline of a “common currency theory” (CCT). A theory is more comprehensive and complex than a hypothesis. We want to point out four core features of a theory, the CCT, which warrants future development. Among others, a theory can be characterized by four features: systematic relationships, mechanisms, models, and predictions; these shall be briefly illustrated in the following.

First, a theory is characterized by systematic relationships among its different elements, features or variables. In the case of the CCT, we assume systematic relationship of neural and mental features through their shared temporo-spatial dynamics. This is supposed to apply to a variety of distinct mental features including self, consciousness, thought, dreams, and various others as described in the empirical sections of this paper (see also [47]). Despite their differences, the various mental features are characterized by certain temporo-spatial dynamical characteristics that they share with the temporo-spatial features of the brain’s neural activity as their “common currency” (see also [47]). One can thus speak of temporo-spatial relationships that characterize different neuro-mental relations in distinct ways.

Second, these systemic relationships allow one to postulate specific mechanisms regarding how one variable is strongly related to another one. For instance, the duration of the brain’s autocorrelation window may impact the balance of temporal integration and segregation of different inputs or stimuli which, in turn, may be manifest in a more fine- or coarse-grained perception – temporal integration may thus be a key mechanism here that connects neural and mental features [92,95]. While further investigation is warranted into the exact mechanism connecting neural and mental states, the CCT claims that these are explicitly temporo-spatial mechanisms (as distinguished from cognitive, affective, sensory, informational, predictive, etc. mechanisms).

5.3. Shifting from “Common currency hypothesis” (CCH) to “Common currency theory” (CCT) ii – integrated brain-mind model and temporo-spatial predictions

Third, the systematic relationships among the different temporal (dynamic) and spatial (topographic) variables together with the temporo-spatial mechanisms connecting neural and mental features allow us to develop a model of neuro-mental relationship. Roughly, a model can be regarded as a virtual realization of the systematic relationships among different variables, that is, those of interest. In the case of the CCT, we aim to develop a model of neuro-mental relationships which focuses on their shared temporo-spatial dynamics as “common currency”. CCT thus proposes an explicitly spatiotemporal model of mental features that needs to be distinguished from other models, which serve as a theoretical and empirical background for investigating and understanding specific neural and mental features. For example, cognitive models of brain and mind provide the framework for a number of current theories of consciousness, such as Attention Schema Theory, Higher-Order Thought theories, and Global Neuronal Workspace Theory [72]. While the information model serves as theoretical framework for the Integrated Information Theory [80]. In that vein, CCT inspires and provides a different theoretical framework, one that proposes an explicitly temporal-spatial model of neuro-mental relationship which, empirically, is applied to different mental features like meditation [9], dreams [57], self [51], consciousness [54,64], thought [47], and mental disorders like depression [53] and schizophrenia [2,94].

Note that CCT does not only provide a model of the mind or mental features. Extending beyond mind models (and also brain models), it aims to provide an integrated brain-mind model, a spatiotemporal model of the brain that includes its connection to the mind. Such an integrated brain-mind model is proposed by the ‘Dynamic layer model of brain’ (DLB) where the brain is characterized by its different temporo-spatial layers as distinguished by their distinct time-space scales, that is, by their distinct spatial extensions and temporal durations, i.e., timescales [47]. The DLB is an integrated brain-mind model that must be distinguished from both current brain models [3,12,67] and mind models [65] (see [47] for details).

Finally, and fourth, a model allows one to make and test specific predictions of the relationships among the distinct variables. One can make the prediction from experiential analysis and spatiotemporal characteristics of mental features to the brain’s neural dynamics and topography, or, reversely, from the analysis of neural dynamics and topography to mental features. For instance, from the experience of the visual environment as too fast for a depressed patient, one may predict that her/his neural activity in visual cortex may be too slow [56,74]. Conversely, one may observe very long autocorrelation windows in the brain of a subject; from there one may predict that subject’s experience or perception may be less fine-grained with lower degrees of temporal segregation and, conversely increased temporal integration of the different stimuli or inputs [92,95]. One can thus speak of temporo-spatial predictions from either neural to mental features or, conversely, from mental to neural features.

Together, these four features (among others), systematic relationships, mechanisms, models, and predictions characterize a theory. The findings in the last four years and the theoretical developments in this and other publications [47] may allow us to transition from hypothesis to theory- that is, from CCH to CCT. CCT provides a highly promising theoretical and empirical framework to understand and investigate the neuro-mental relationship in a novel way. This is illustrated in our 2020 paper and by the recent progress made on the temporo-spatial characterization of various mental features - which we describe in the current paper and a recent book [47] - as well as by the fruitful application of CCT in other domains like psychiatry [48]. However, future empirical, computational and mathematical development is certainly necessary to develop a full-blown spatiotemporally based and integrated brain-mind theory of mental features.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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